Weather Data Analysis & Next-Day Temperature Prediction

Time-series regression project report

1. Objective & data

The goal is to analyse historical weather trends and predict **tomorrow's mean temperature** from today's meteorological readings, recent history and calendar features. The dataset holds 4,018 daily records (01 Jan 2015 to 31 Dec 2025) with temperature, humidity, wind speed, pressure and precipitation.

Data note: this run uses a **synthetic dataset** calibrated to a subtropical monsoon climate (hot summers, monsoon rain from late June to September, mild winters, slow warming trend). It was generated because the build environment had no internet access. The pipeline reads a standard CSV, so real station data (NOAA, Meteostat, Kaggle) can be dropped in without code changes.

2. Cleaning summary

Step	Result
Raw rows	4,033
Duplicate rows removed	15
Missing cells before	322
Outliers flagged (rolling-median filter)	30
Missing cells after time interpolation	0
Clean daily rows	4,018

3. Exploratory data analysis

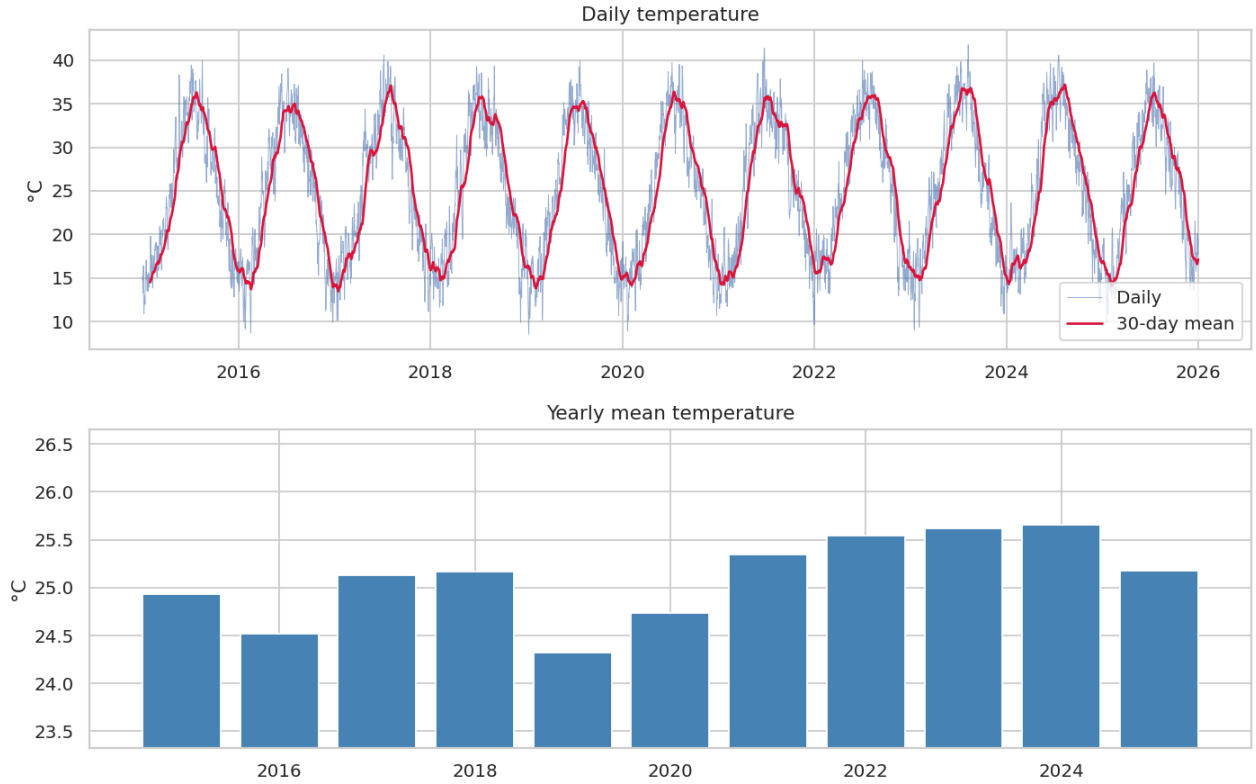


Fig 1. Daily temperature with 30-day mean, and yearly averages showing a mild upward trend.

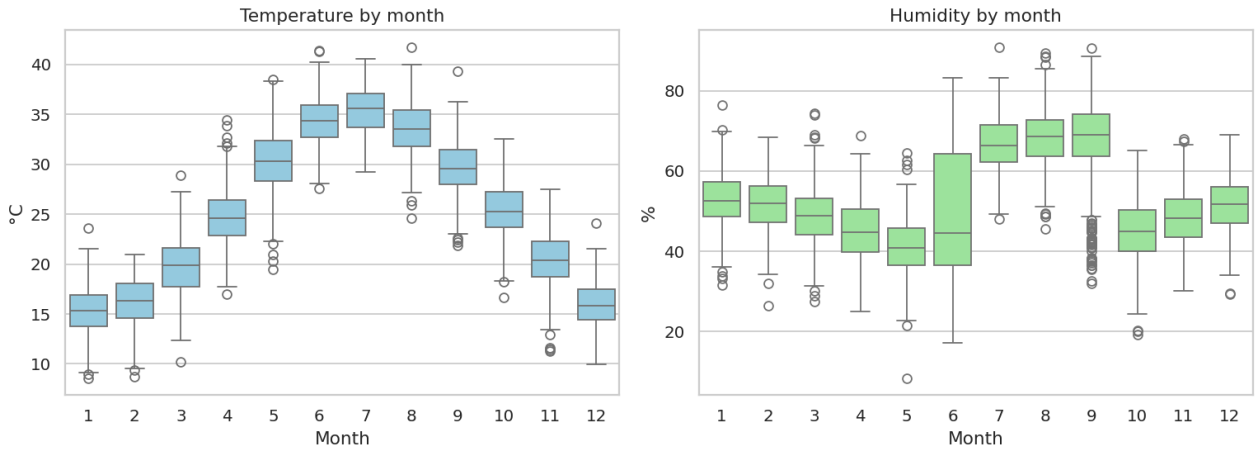


Fig 2. Clear seasonality: temperature peaks May to June, humidity jumps during the monsoon.

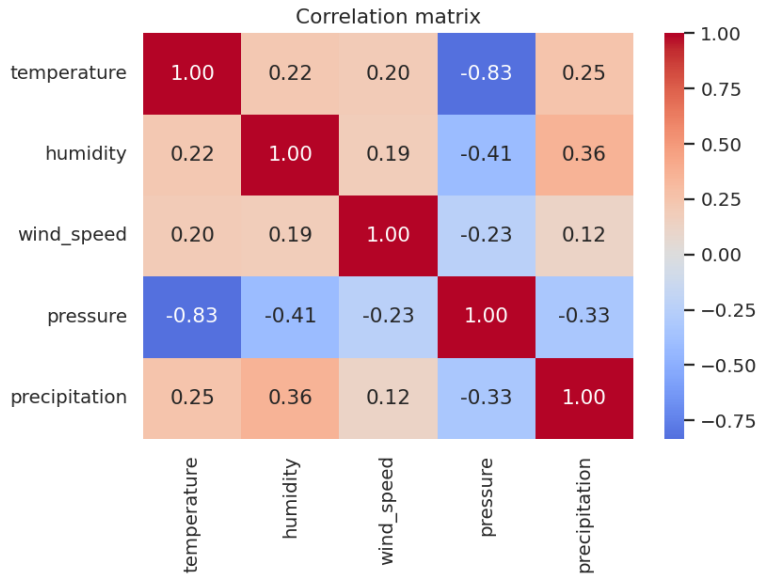


Fig 3. Correlations. Humidity and pressure move against temperature; wind and rain are weak linear predictors.

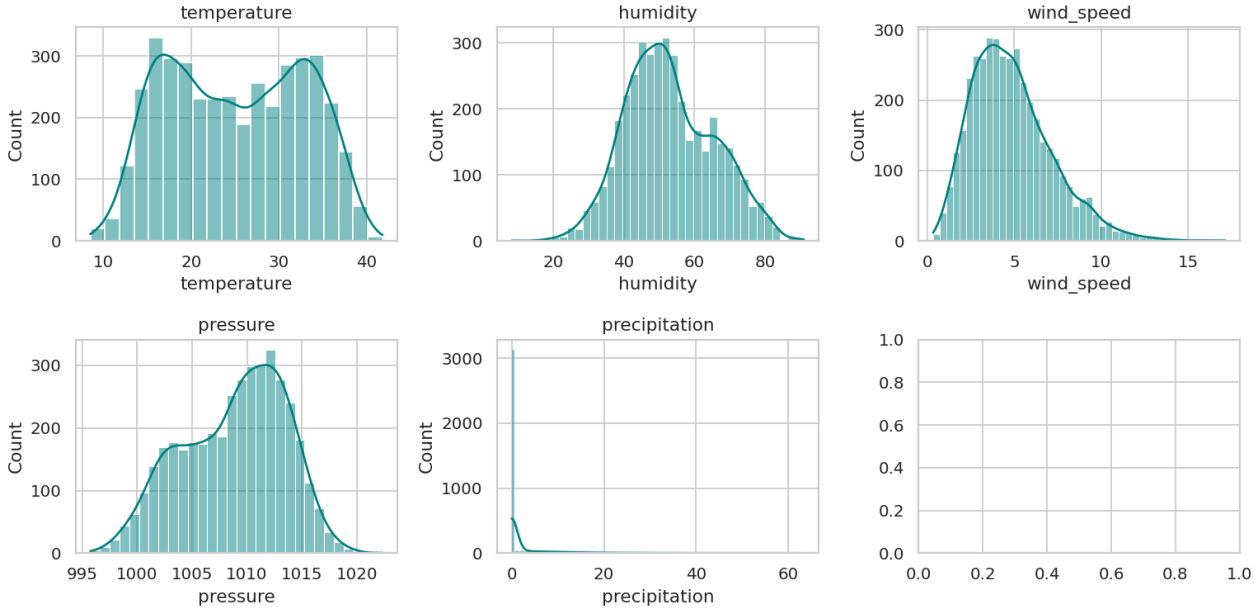


Fig 4. Feature distributions (post-cleaning). Wind and precipitation are right-skewed.

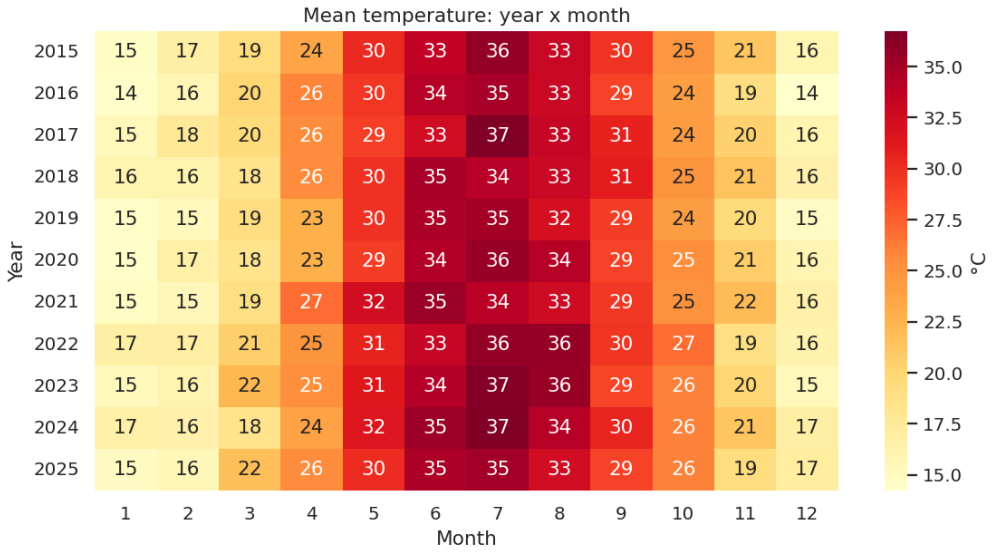


Fig 5. Year x month mean temperature.

4. Feature engineering & split

Features: calendar variables (year, month, day-of-year with sine/cosine encoding, season dummies); temperature lags (1, 2, 3, 7, 14 days); rolling means (3, 7, 30 days) and 7-day std; humidity and pressure lags, pressure change, 7-day rainfall. All features use only data up to day *t* to predict day *t+1*. The split is chronological: train 2015-01-30 to 2023-10-24 (3,190 days, 80%), test 2023-10-25 to 2025-12-30 (798 days, 20%). No shuffling, so there is no leakage from the future.

5. Model comparison

Model	MSE	RMSE (°C)	MAE (°C)	R^2
Persistence (naive)	3.296	1.815	1.450	0.9440
Linear Regression	2.841	1.686	1.344	0.9517
Ridge (CV alpha)	2.843	1.686	1.344	0.9517
Random Forest	3.036	1.742	1.393	0.9484

Best model by RMSE: **Linear Regression**. The persistence row (tomorrow = today) is the sanity benchmark every model must beat.

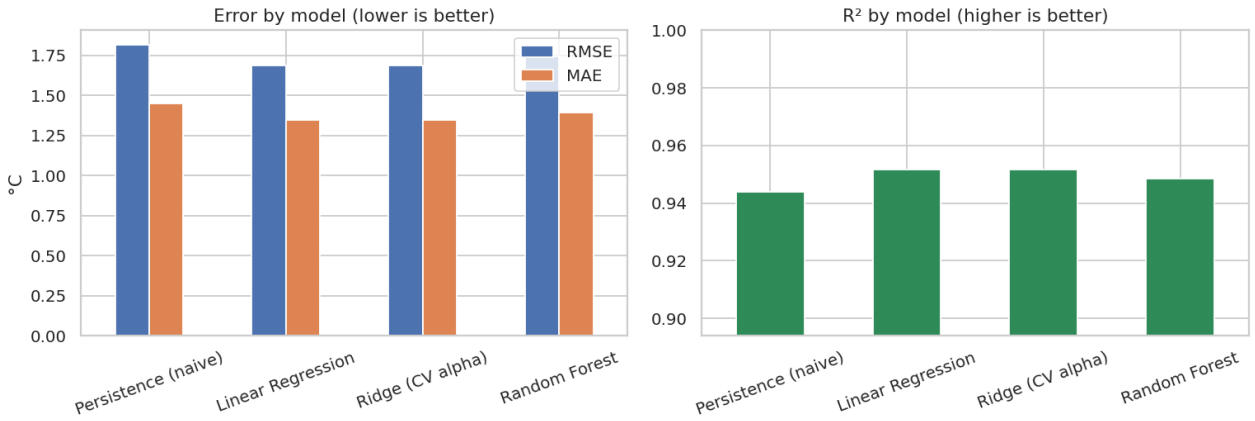


Fig 6. Error and R^2 comparison.

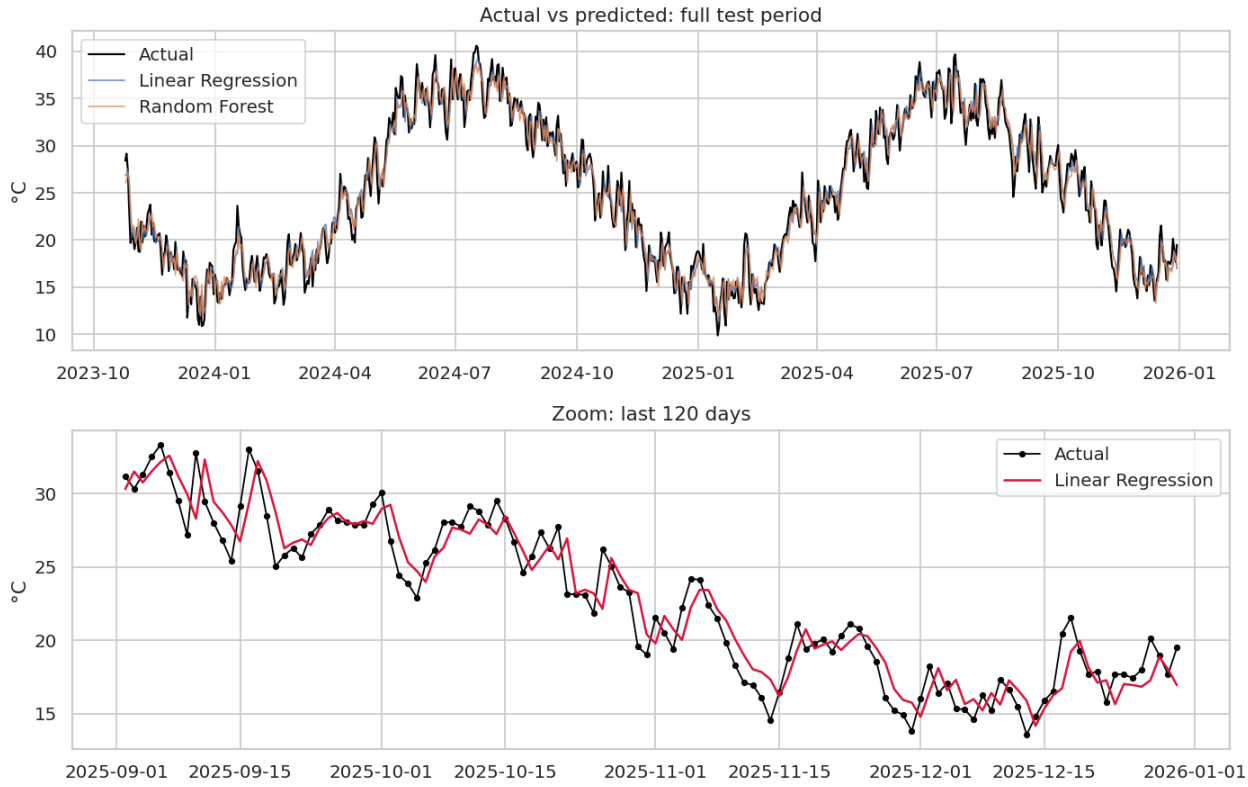


Fig 7. Actual vs predicted on the held-out test period.

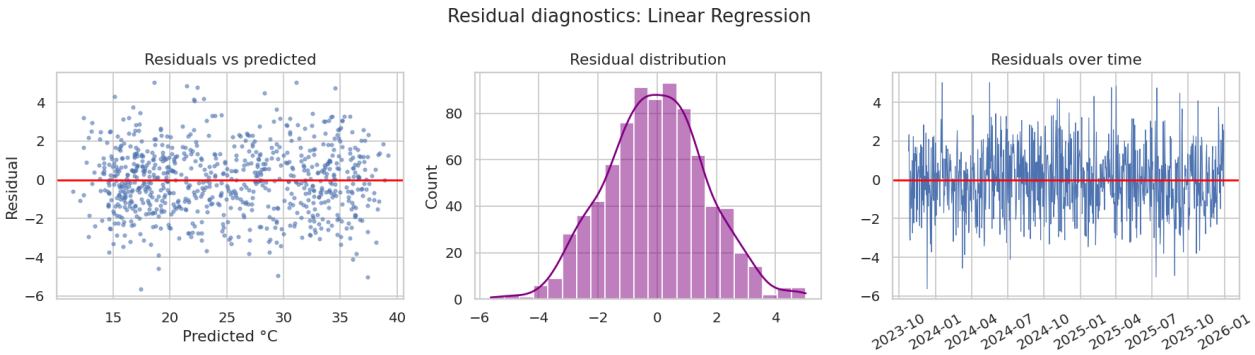


Fig 8. Residuals: centred on zero with no strong pattern suggests little systematic bias.

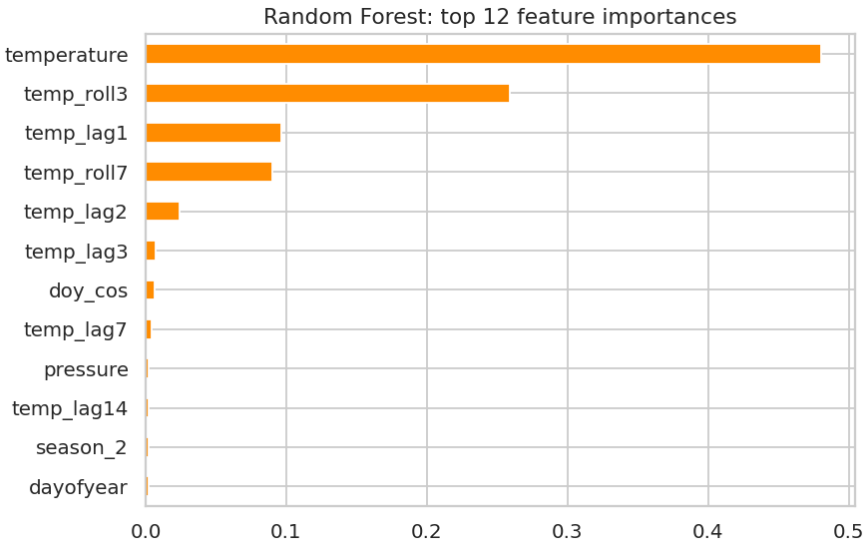


Fig 9. Random Forest feature importances.

6. Conclusions & next steps

• Lag and rolling-mean features carry most of the signal; weather is strongly autocorrelated.
• Regularised linear models perform on par with Random Forest here because the seasonal relationship is largely linear once encoded cyclically.
• Next steps: try Gradient Boosting / XGBoost, multi-day horizons, walk-forward validation, and run on real station data where non-linear effects (fronts, ENSO) should favour the tree models more.
• Caveat: results on synthetic data demonstrate the workflow, not real-world forecast skill.